Research Memo 06: Standards, Test Requirements, and Certification Scope

> ### UPDATED 18 Sep 2026 — the certificate is now a specific number
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> The engine's rating plate has been photographed. Two things change for this memo, both in the direction of making its conclusions **more** concrete, not less:
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> - The engine is **3GK550ETA 4SR1**, application code `GK3.8703`, not the `3R550ETA 4G1` this memo inherited from research memo 01. The genset model `KG4-25WS1` in the header was inferred from the same chain and should be treated as **unconfirmed** — no genset nameplate has been seen.
> - The plate carries the actual type approval: **`ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`**. `IGES4` is India Genset Emission Standard IV, confirming the CPCB IV+ tier from the machine itself rather than from a brochure.
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> **This memo's regulatory conclusions stand, and now attach to a numbered certificate against a specific serial number** rather than to a product family in the abstract. That is worth stating plainly to whoever takes the decision.
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> One gap this memo did not examine, raised by a later review: the **state pollution board Consent-to-Operate** regime is a separate gate from central type approval and was never looked at. Recorded as unexamined.
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> The controller references below to the KG640C are also superseded — the fitted unit is a **DSE4522 MKII AMF**. See `refs/field-evidence-2026-09-18.md`.

Task: U6 (J1939 layers, ISO 7637-2 transient immunity, CISPR 25 conducted emissions, certification position). Consumes Task 1's emissions tier. Produces the test/design numbers Phase 3 layout must honour and an honest scope statement on certification. **Date:** 2026-09-13 **Author:** Phase-1 research (task 6) **Genset:** Kirloskar KG4-25WS1, 25 kVA, 415 V 3-phase / 230 V 1-phase, 50 Hz. **Engine:** Kirloskar 3R550ETA 4G1, 26.5 kW prime (19–56 kW CPCB IV+ band). **Emissions tier:** CPCB IV+, cooled EGR + DOC, no SCR/DEF/DPF (per `docs/research/01-engine-identity.md`). 12 V electrical system. In service, in India.

Verdict (read this first)

Item	Finding	Status
J1939 physical/transport/diagnostic layers	Standard 250 kbit/s twisted-pair CAN with 120 Ω termination, 1 m stub limit; BAM for broadcast multi-packet, RTS/CTS for destination-specific; DM1/DM2 for active/previously-active DTCs; address-claim on power-up	Secondary-sourced (SAE texts paywalled); PGN set itself carried over from Task 2, there marked unconfirmed for this specific controller
ISO 7637-2 transient levels (12 V ref.)	Pulse 1: −150 V (Level IV); Pulse 2a: +112 V; Pulse 2b: +10 V; Pulse 3a: −220 V; Pulse 3b: +150 V (2 ms–0.1 ms widths, 2–50 Ω source impedance per pulse); Pulse 5b (load dump, now properly an ISO 16750-2 pulse, still colloquially cited under 7637-2): 40–105 V suppressed, 0.5–8 Ω, 40–400 ms	Secondary source (industry technical note quoting the standard's table), clearly marked — standard text itself not accessed
CISPR 25 conducted emissions	Class 5 (most stringent), 150 kHz–108 MHz, voltage method via 5 µH/50 Ω LISN; toughest bands are VHF/FM (68–108 MHz): 18 dBµV avg / 38 dBµV peak	Secondary source, clearly marked
Certification position	**Confirmed, from the gazette text itself:** CPCB IV+ type approval is granted to an engine/genset family and obtained by the manufacturer/importer/assembler at the point of manufacture or import — it is not a per-unit approval that travels indefinitely with the physical genset regardless of what is later done to it. **A separate CPCB regulatory document (the RECD System & Procedure) states directly that ECU characteristics governing injection timing, air-mass metering, and emission-reduction strategy must not be altered even under a government-sanctioned retrofit programme, and that modifications limited to engine-control changes fall outside the one retrofit-approval pathway CPCB operates.** No pathway was found by which a third-party ECU replacement on this genset could be approved, and General Condition 3 of GSR 804(E) states plainly that no person shall use a genset lacking a valid Type Approval + CoP certificate. **This is a decision for the project owners to take to qualified counsel and/or CPCB/ARAI directly — not an engineering call, and not something this memo resolves.**	Confirmed (primary gazette text and primary CPCB procedure text, both fetched and read in full) for what the rules say; **not legal advice** on what they mean for this specific project

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Step 1: SAE J1939 layers that apply

SAE's J1939 series texts are paywalled (sae.org); nothing below was read from the standard itself. Every figure is from secondary industry sources, cited individually, and should be treated as **secondary/summarized**, not as a verbatim standard quote.

- **J1939-11 (physical layer, twisted-pair variant).** Shielded twisted-pair backbone, nominal 120 Ω characteristic impedance, 250 kbit/s data rate for this device class, 120 Ω (±1%, ¼ W typical) termination resistors at both physical ends of the backbone, and stub ("drop line") length limited to ≈1 m per node to control reflections. *(Secondary: https://simmasoftware.com/products/j1939/sae-j1939-11-physical-layer/ ;*

https://copperhilltech.com/content/SAE%20J1939%20Wiring%20Requirements.pdf ; https://obd-cable.com/j1939-backbone-termination-stub-length-guide/.) **Design relevance:** the genset's CAN segment (GCU↔ECU, per Task 2's KG640C memo) should be laid out to this backbone/stub/termination discipline; a single genset with two nodes is a short, simple topology relative to the standard's truck/trailer use case, so termination is the only piece likely to bite if omitted.

- **J1939-21 (transport protocol).** Multi-packet (>8 byte) messages use either **BAM** (Broadcast Announce Message — connectionless, one-to-all, no flow control, used for messages any node may want, e.g. some diagnostic broadcasts) or **RTS/CTS** (Request-to-Send/Clear-to-Send — connection-oriented, destination-specific, flow-controlled, used for point-to-point large transfers). *(Secondary: https://www.spcontroller.com/j1939-transport-protocol-bam-vs-rts-cts/ ; https://copperhilltech.com/blog/sae-j1939-programming-with-arduino-transport-protocol-rtscts-session-sae-j193921/.)* **Design relevance:** if the replacement ECU needs to send anything larger than a single CAN frame (e.g., a multi-DTC DM1 payload) to the GCU, the firmware needs both mechanisms implemented — most stacks default to BAM for DM1 and would only need RTS/CTS if a proprietary large-payload message is added.
- **J1939-71 (application layer / PGN definitions).** Task 2 (`docs/research/02-kg640c-gcu.md`, §2) already established the working PGN set for this specific genset, **explicitly flagged there as an inferred industry-baseline, not confirmed against a KG640C document**: EEC1 (PGN 61444, engine speed), ET1 (PGN 65262, coolant temp), EFL/P1 (PGN 65263, oil pressure), VEP1 (PGN 65271, battery/system voltage), DM1 (PGN 65226, active DTCs), and TSC1 (PGN 0, torque/speed control, if the GCU is to command the engine). This memo does not re-derive that list; it is carried forward as-is, with the same caveat Task 2 attached.
- **J1939-73 (diagnostics, DM1/DM2). DM1 — Active Diagnostic Trouble Codes** (PGN 65226/0xFECA): broadcast by default at 1 Hz and on state change, carries currently-active DTCs. **DM2 — Previously Active Diagnostic Trouble Codes** (PGN 65227): available on request only, carries DTCs that were active but have cleared. *(Secondary: https://www.csselectronics.com/pages/j1939-73-dm1-diagnostic-message-dtc.)* **Design relevance:** the replacement ECU should broadcast DM1 unsolicited at 1 Hz to be a drop-in match for what the KG640C GCU's fault-display logic almost certainly expects (per Task 2's finding that the GCU is a CAN consumer, not just a display).
- **J1939-81 (network management / address claiming).** On power-up, each node transmits an Address Claimed message; every other node records/verifies it; if two nodes claim the same address, priority is resolved by the NAME field (a 64-bit value encoding manufacturer, function, and an arbitrary/self-configurable-address capability bit). *(Secondary: https://copperhilltech.com/blog/sae-j1939-address-claim-procedure-sae-j193981-network-management/.)* **Design relevance:** the

engine ECU node conventionally claims source address 0x00 by industry convention (again per Task 2, unconfirmed for this controller specifically) — the replacement ECU should claim the same address the GCU expects, which direct bus logging (Task 2's recommended next step) would settle definitively.

Net for Phase 3 layout: none of this drives component values the way Steps 2–3 do; it drives firmware behaviour and connector/stub layout. The one open item carried from Task 2 — the real PGN set and source address, unconfirmed from any KG640C-specific document — should be closed by logging the existing bus before the replacement ECU's CAN stack is finalized.

Step 2: ISO 7637-2 transient immunity — the numbers that set the TVS and buck input rating

Source note: the ISO 7637-2 standard text itself is paywalled and was not accessed. The table below is drawn from an ON Semiconductor automotive applications technical note that reproduces the ISO 7637-2:2011(E) severity-level table directly (title "Supplemental Data for ISO7637 Pulses," https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf, fetched and text-extracted in full) — **marked here as secondary**, since it is an industry document quoting the standard, not the standard.

Pulse	Description	Amplitude, 13.5 V nominal system, Level I/II	Level III	Level IV	Source impedance / pulse width	Test duration
1	Supply disconnection (inductive load switch-off, e.g. relay coil)	−75 V	−112 V	**−150 V**	10 Ω, 2 ms	500 pulses, min. 0.5 s repeat
2a	Sudden interruption of current in a device sharing the harness with the DUT (inductive)	+37 V	+55 V	**+112 V**	2 Ω, 0.05 ms	500 pulses, min. 0.2 s repeat
2b	Same mechanism, lower-energy variant	+10 V	+10 V	+10 V	2 Ω, 0.05 ms	10 pulses, min. 0.5 s repeat
3a	Fast transient bursts (switching-type, EFT-like)	−112 V	−165 V	**−220 V**	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat
3b	Same mechanism, opposite polarity	+75 V	+112 V	**+150 V**	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat

(All figures per ON Semiconductor TND6424-D, secondary source.)

Load dump (Pulse 5b) — a nuance worth stating plainly: the 2011 edition of ISO 7637-2 removed the load-dump pulses (5a/5b); they now live in **ISO 16750-2** (and are referenced in ISO 21848 for some classes), though industry practice — including many component datasheets and this project's own brief — still refers to "ISO 7637-2 pulse 5b" as shorthand. *(Secondary: search-result synthesis, not independently verified against either standard's text — flagged as an open item rather* *than a confirmed edition history. If the design team needs to cite the pulse in a test plan, ISO 16750-2 §4.6.4 is the more defensible current reference.)* For a **suppressed** load dump (the realistic case on a genset with an alternator-integrated or centralized suppression diode/clamp, as is now common) — pulse 5b: peak voltage 40–105 V for a 12 V system, source impedance 0.5–8 Ω, pulse width 40–400 ms. *(Secondary, search-result synthesis — not read from either standard's own table.)*

What this sets for Phase 2/3 hardware: - **TVS clamp voltage:** must sit above the normal 12 V (13.5–14.4 V charging) rail with margin, but clamp well below the injector boost-converter's and any other sensitive node's absolute maximum — the working design window is a clamp that lets pulses 2a/2b (+37 to +112 V) and suppressed 5b (up to ~105 V) pass through to a controlled clamp voltage, while pulses 1/3a (down to −220 V) are clamped on the negative side. A single automotive-grade bidirectional TVS rated for these levels (many parts are explicitly binned "meets ISO 7637-2 pulse 1/2a/3a/3b," per component datasheets — not independently verified against this list here) is the usual answer; the −220 V pulse 3a figure (Level IV, 50 Ω/0.1 ms — low energy despite high voltage) is the most demanding negative excursion to clamp. - **Buck converter input rating:** the input stage (from battery rail to the point after transient protection) should be rated for a working input up to the clamped level of the worst transient the TVS lets through (commonly designed to ~40–45 V clamped for a suppressed load-dump system, consistent with the Vishay/EDN load-dump note found in this search pass but not independently verified here) rather than the full unclamped 220 V pulse, which the TVS/clamp stage is responsible for absorbing before it reaches the buck's input pins.

Step 3: CISPR 25 conducted emissions — band limits and what a bench setup can tell you

Source note: the CISPR 25 standard text itself is paywalled and was not accessed. Figures below are from two secondary sources: a Texas Instruments technical article (https://www.ti.com/lit/pdf/sszt673, fetched and read in full) and an EMC-limits compilation site (https://emccalc.com/limits/cispr25-cev/, fetched). **Both are marked secondary — neither is the standard itself.**

- **Method and frequency range:** conducted emissions are measured at the DUT's power terminals via a 5 μH/50 Ω LISN (line impedance stabilization network), voltage method, over **150 kHz to 108 MHz**, using peak and average detectors. *(TI, sszt673, confirmed as stated by that source.)*
- **Class 5** is the most stringent of CISPR 25's five classes and the one automotive OEMs typically mandate for safety-/receiver-adjacent electronics. *(TI, sszt673.)* Selected Class 5 limits, voltage method (secondary, per emccalc.com's compilation, which states the 2016 (Ed.4) and 2021 (Ed.5) voltage-method limits are identical):

Band	Frequency	Peak limit	Average limit
LW (broadcast)	0.15–0.30 MHz	70 dBµV	50 dBµV
MW (broadcast)	0.53–1.8 MHz	54 dBµV	34 dBµV
SW (broadcast)	5.9–6.2 MHz	53 dBµV	33 dBµV
CB	26–28 MHz	44 dBµV	24 dBµV
VHF (low)	30–54 MHz	44 dBµV	24 dBµV
VHF (high)	68–87 MHz	38 dBµV	18 dBµV
FM broadcast	76–108 MHz (overlapping VHF-high per source)	38 dBµV	18 dBµV

The **68–108 MHz VHF/FM region carries the tightest limit (18 dBµV average / 38 dBµV peak)** — explicitly called out by TI as "extremely challenging." *(TI, sszt673, confirmed as stated by that source.)* This is the band a switching converter's harmonics most often violate, and it is where PCB layout (input filter corner frequency, output ripple, switching-edge control) earns its keep.

What a LISN + spectrum analyser bench setup can and cannot tell you, short of a test house: - **Can:** give a reliable relative measurement of your own converter's conducted noise floor and harmonic content across the 150 kHz–108 MHz range, using a 5 µH/50 Ω LISN (buildable or purchasable as a discrete unit) feeding a spectrum analyser with peak-hold — enough to catch a converter that is grossly non-compliant (wrong switching frequency choice, missing input filter, poor grounding) long before a certification test, and to A/B compare layout or filter revisions against each other. - **Cannot:** substitute for accredited peak/average/quasi-peak detector calibration, the specific bonded ground-plane/test-bench geometry CISPR 25 mandates, cable harness length and dress requirements, or the formal measurement uncertainty budget a test house applies — none of which a bench LISN setup replicates exactly. A bench pass is **necessary-but-not-sufficient** evidence; it de-risks the design before a paid test-house slot, it does not replace one. *(This paragraph is general EMC test-practice knowledge, not tied to a specific citation found in this search pass — flagged as such rather than presented as sourced.)*

Step 4: The certification position — stated plainly

This is the section the brief calls the one that matters most, and it is written to stand on its own.

4.1 What CPCB IV+ actually is, and where it comes from — confirmed from the gazette text itself

This memo obtained and read, in full, the actual Gazette of India notification: **G.S.R. 804(E), Ministry of Environment, Forest and Climate Change, New Delhi, 3 November 2022** — the *Environment (Protection) Third Amendment Rules, 2022*, made under **sections 6 and 25 of the Environment (Protection) Act, 1986**, amending **Schedule I, serial number 88** of the Environment (Protection) Rules, 1986. *(Confirmed — primary source, fetched directly at* *https://cpcb.gov.in/openpdffile.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNl9tZWRpYXBob3RvMTkwNTgucGRm and text-extracted via `pdftotext -layout` from the saved PDF; this upgrades Task 1's "inferred/secondary" GSR citation to confirmed.)* It comes into force **1 July 2023**, exactly as Task 1 recorded.

The band limits are now confirmed directly from the gazette's own Table 1, not from dieselnet's summary as Task 1 had it: for the **19 < P ≤ 56 kW** power category (this engine's 26.5 kW prime rating falls here) — combined **NOx+HC ≤ 4.7 g/kWh, CO ≤ 3.5 g/kWh, PM ≤ 0.03 g/kWh, smoke ≤ 0.7 m^{-1}** (light absorption coefficient), for CI engines, no separate NOx-only limit in this band. This matches dieselnet's figures exactly and is now **confirmed**, not inferred.

4.2 What the certification covers, and who holds the obligation — confirmed from the gazette's General Conditions

- **General Condition 1 (Applicability):** the rules apply to "**new** engines for power generation application and products **manufactured, assembled or imported to India**" — this is a point-of-manufacture/import regime, not a running in-service inspection scheme (there is no equivalent here to a vehicle's periodic PUC check). *(Confirmed, direct gazette quote.)*
- **General Condition 2 (Requirement of certification): "Domestic manufacturer, importer or assembler**... shall obtain **Type Approval** from authorised certifying agency and also comply with **Conformity of Production (CoP)** test of their product(s)..., valid for the next Conformity of Production year." *(Confirmed, direct gazette quote.)* The obligation to *hold* a Type Approval and CoP certificate is placed on the **manufacturer/importer/assembler**, not on the end owner or operator.
- **General Condition 3 (Sale/import/use of non-compliant product):** "**No person shall sell, import or use** an engine and genset for power generation application which is not having a valid Type Approval certificate and certificate of Conformity of Production." *(Confirmed, direct gazette quote — this is the clause that reaches beyond the manufacturer to anyone who* uses *such equipment, which would include a genset owner/operator.)*
- **General Condition 5 (Nodal agency):** the **Central Pollution Control Board (CPCB)** is the nodal agency for implementation. *(Confirmed.)*
- **General Condition 6 (Authorised certifying agencies): Automotive Research Association of India (ARAI, Pune)**, **International Centre for Automotive Technology (ICAT, Manesar)**, and **Indian Institute of Petroleum (IIP, Dehradun)** are the agencies authorised to test and certify Type Approval and CoP for engines and gensets. *(Confirmed, direct gazette quote; a related CPCB procedure document for the separate retrofit-device scheme, §4.2.1, also lists Vehicle Research and Development Establishment (VRDE, Ahmednagar) and any NABL-accredited agency meeting that scheme's requirements — this fourth agency and the NABL route are confirmed only for the retrofit-device scheme, not shown in the GSR 804(E) text itself for new-engine Type Approval.)*

- **General Condition 11:** CPCB, the Commission for Air Quality Management, State Pollution Control Boards, or Pollution Control Committees may issue **more stringent** local norms. *(Confirmed — a reminder that state-level rules could add obligations beyond the national floor; not researched further here, flagged as an adjacent open item.)*

What "Type Approval" covers, structurally: it is granted to an engine/genset **family or model** — the regulation's own "engine family" concept (defined in the companion CPCB retrofit-device procedure, §1.1.18: "a manufacturer's grouping of engine systems, which, through their design, have similar exhaust emission characteristics; all members of the family must comply with the applicable emission limit values," citing ISO 8178-7) is confirmed only in that companion document but is standard practice across this same regulatory architecture (same nodal agency, same three certifying agencies, same GSR-804(E)-referenced testing regime). Individual units are not independently re-tested against Table 1 one-by-one; the manufacturer's ongoing conformance is checked by periodic Conformity of Production (CoP) sampling of production units, not a certificate that attaches permanently, in isolation, to one physical serial number regardless of what happens to it afterward.

4.3 The direct hit: CPCB's own text on ECU/calibration modification

This memo located and read, in full, **CPCB's "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW"** (fetched from https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, text-extracted via `pdftotext -layout` from the saved PDF in full, 3258 lines). This is CPCB's own regulatory document governing the *one* pathway CPCB operates for modifying an already-type-approved genset's emission-control configuration after manufacture. Two passages from it are decisive for this project, quoted verbatim:

> **§4.1.2.2(e):** "The characteristics of the electronic engine control unit (as regards, for example, injection timing, air-mass flow metering, or exhaust emissions reduction strategies) **shall not be altered** by the retrofitting. Any modification of the test engine that changes the original emission behaviour (for example alternation of the injection timing) **is not permitted.**"
>
> **§1.1.35 (definition of "Retrofit emission control device"):** "...Systems that **only modify the existing engine system controls are not considered to be RECDs.**"

(Both confirmed — direct quotes, primary CPCB source, fetched and read in full.)

Read together, these two passages mean: (1) CPCB explicitly identifies **injection timing, air-mass flow metering, and exhaust-emission-reduction strategy** — precisely the calibration parameters this engine's EGR-rate-scheduling-and-injection-timing NOx/PM strategy runs on (per Task 1's finding of EGR+DOC-only aftertreatment, no SCR) — as characteristics that must not be touched, even under a CPCB-sanctioned emissions-improvement retrofit programme; and (2) a modification that is *purely* a change to the engine's own control system, with no added physical after-treatment hardware — which is exactly what an ECU replacement is — **does not qualify as an RECD at all** and therefore has **no approval pathway under this scheme, full stop.** It is neither covered by the new-manufacture Type Approval/CoP scheme (GSR 804(E), which governs manufacturers at point of production/import, §4.2 above), nor by the one modification-approval scheme CPCB operates for in-service units (RECD, which by its own definition excludes pure control-system changes).

Negative result, recorded per the brief's instruction: no CPCB-administered, or any other, approval pathway was found in this search by which replacing this genset's OEM ECU with a third-party unit of different calibration could be formally re-certified or otherwise sanctioned as compliant. This does not prove no such pathway exists anywhere in Indian regulatory practice — only that this search, within its budget, did not locate one. This is exactly the kind of fact pattern that should go to CPCB, ARAI, or qualified counsel directly, not be inferred further by this memo.

4.4 The calibration angle the brief asked to check — confirmed as directly relevant, not merely plausible

The brief's suspicion is correct, and the gazette text itself confirms the mechanism: GSR 804(E) Table 1 sets limits in **g/kWh over a defined test cycle** (a mass-of-pollutant-per-unit-of-work measure), not as a hardware bill-of-materials requirement. Compliance is therefore inherently a function of *how the engine is run* — injection timing, EGR rate scheduling, fuelling — as much as *what parts are bolted to it.* Combined with §4.3's finding that CPCB itself names injection timing, air-mass metering, and emission-reduction strategy as the calibration parameters that determine emission behaviour, and with Task 1's finding that this specific engine meets its 19–56 kW-band NOx+HC limit (≤4.7 g/kWh) through cooled EGR and DOC alone, with **no SCR** as a chemical backstop: **a replacement ECU with different EGR-scheduling or injection-timing calibration can move this engine out of compliance with its certified emission limits without any hardware changing at all.** This is not a remote or theoretical risk — it is the same mechanism by which the engine meets its limits in the first place, run in the wrong direction. It is directly relevant to, and reinforces, the calibration-data gap already flagged as the project's largest single risk in `docs/research/04-injector-drive.md` (Step 5): that memo's finding was about getting the engine to run *correctly and safely* without OEM fuelling maps; this memo's finding is that even a "correctly running" replacement ECU, if its EGR/injection-timing calibration differs from the certified original, has no assurance of remaining within the CPCB IV+ limits the genset was certified against — an emissions-compliance risk layered on top of the combustion/mechanical risk Task 4 already documented.

4.5 Statutory exposure — flagged, not advised on

Separately from the certification-scheme mechanics above, the **Environment (Protection) Act, 1986, Section 15** provides a general penalty for contravention of the Act or rules made under it. *(Secondary — this memo relied on a legal-database rendering of the statute, https://indiankanoon.org/doc/24023684/, rather than a freshly fetched Gazette copy of the Act itself, and the search result returned text that appears to blend the original 1986 penalty (imprisonment up to 5 years and/or fine up to ₹1 lakh, escalating for continuing contravention) with a later monetary-penalty band (₹10,000–₹15 lakh) that reads as a post-2023 Jan Vishwas (Amendment of Provisions) Act amendment. Which version is currently in force, and whether it would apply to this specific fact pattern, is exactly the kind of question that needs a lawyer, not this memo — flagged, not resolved.)*

4.6 The decision, framed as the brief asked

This memo does not tell the project owners whether to proceed. What it establishes, with the citations above:

1. CPCB IV+ Type Approval is a family/model-level certificate obtained by the manufacturer, not a permanent per-unit status immune to what happens to the physical genset afterward.
2. The obligation to *hold* Type Approval + CoP sits with the manufacturer/importer/assembler; the obligation not to *use* a non-compliant engine/genset (General Condition 3) reaches the owner/operator directly, by the gazette's own words — "no person shall... use."
3. CPCB's own regulatory text, in the one scheme it operates for modifying in-service gensets, states plainly that altering ECU characteristics governing injection timing and emission-reduction strategy is not permitted even under a sanctioned retrofit programme, and that pure engine-control modifications fall outside that scheme's approval process entirely.
4. No approval pathway for a third-party ECU replacement on this genset was found in this search.
5. The mechanism by which this specific engine meets its CPCB IV+ limits (EGR scheduling + injection timing, no SCR backstop) is exactly the mechanism a replacement ECU's calibration would change.

Whether this means the genset's certification is void, whether it creates a compliance or disclosure obligation, and what exposure attaches to the owner/operator versus this project team, is a legal and regulatory question — not an engineering one. It should go to CPCB,

one of the three authorised certifying agencies (ARAI/ICAT/IIP), or qualified counsel, before this project proceeds past the bench-verification stage Task 4 already scoped as safe to continue.

Sources consulted

- `/home/mohan/workws/ecu25kva/.superpowers/sdd/2026-09-13-phase1-research/task-6-brief.md` — task requirements, read in full.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine identity and emissions-tier memo, read in full (Step 4 specifically).
- `/home/mohan/workws/ecu25kva/docs/research/02-kg640c-gcu.md` — KG640C GCU/J1939 memo, grepped for PGN references (Step 2 of that memo).
- `/home/mohan/workws/ecu25kva/docs/research/04-injector-drive.md` — injector drive/calibration-boundary memo, read in full (Step 5 specifically, for the calibration-risk cross-reference in §4.4 above).
- **G.S.R. 804(E), 3 November 2022**, Ministry of Environment, Forest and Climate Change — fetched directly (https://cpcb.gov.in/openpdffile.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNl9tZWRpYXBob3RvMTkwNTgucGRm), saved and text-extracted in full via `pdftotext -layout`. **Primary source, read in full.** This is the actual Gazette of India notification text — confirms GSR number, date, ministry, Table 1/Table 2 emission limits, and all General Conditions cited in Step 4.
- **CPCB, "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW"** — fetched directly (https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, redirected from cpcb.nic.in), saved and text-extracted in full via `pdftotext -layout` (3258 lines). **Primary source, read in full** for the definitions and §4.1.2.2(e) quoted in Step 4.
- https://cpcb.gov.in/genset-notifications/ — fetched, confirms the GSR 804(E)/GSR 436(E) amendment chain exists on CPCB's own notification index.
- https://indiankanoon.org/doc/24023684/ (via search) — Environment (Protection) Act 1986, Section 15 text, secondary rendering of the statute, not independently fetched from a gazette copy of the Act.
- https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf — ON Semiconductor technical note reproducing the ISO 7637-2:2011(E) pulse severity table; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source.
- https://www.ti.com/lit/pdf/sszt673 — Texas Instruments, "A Review of EMI Standards, Part 1 – Conducted Emissions"; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source, for CISPR 25 method and Class 5 VHF/FM limits.

- https://emccalc.com/limits/cispr25-cev/ — CISPR 25 Class 5 voltage-method limit table compilation; fetched. Secondary source.
- https://www.araiindia.com/certification/certification/genset ; https://cpcb.nic.in/type-approval-cop-data/ ; https://cpcb.nic.in/displaypdf.php?id=Tm9pc2VfYW5kX0dlbnNldHMvU3lzX1Byb2RfZm9yX0VtaXNzX0NvbXBfb2ZfREdTZXRzLnBkZg%3D%3D — found via search, corroborate the general Type Approval/CoP scheme description in §4.2 without being individually quoted.
- Searches returning only partial or negative results (recorded, not omitted): `ISO 7637-2 pulse 1 2a 2b 3a 3b 5b amplitude 12V test levels table` (general web search did not surface a full table directly, resolved via the onsemi PDF instead); `CISPR 25 class 5 limits table dBuV band I II III IV V` (resolved via emccalc.com); direct WebFetch of https://keystonecompliance.com/iso-7637-2/ and https://www.electronicdesign.com/.../cispr-25-class-5... returned only qualitative descriptions, no numeric tables — superseded by the sources above.